

Question number	Answer	Additional guidance	Mark
8(a)	An answer that includes three of the following points: - mitochondria {damaged / destroyed/reduced in number} (by ROS) (fewer mitochondria) results in reduced energy production(1) - damage to {nerve cells/neurones /synapses / dendrites}(1) - named change in brain chemistry(1)	accept {increase in / accumulation of} {ROS/ oxidative stress/peroxide/calcium} aging causes changes in the brain {size / blood flow} accept reduced {nerve impulses/depolarization} ignore nerve tissues / neurodegeneration eg. loss of {dopamine/serotonin}	(3)

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8(b)	An explanation that includes three of the following points: - hydrogen (atoms from reduced co-enzymes) split into {H⁺ / protons} and electrons (1) (electrons release) energy to pump {H⁺ / protons} into intermembrane space (1) {protons / hydrogen ions} move through {membrane protein/ ATP synthase/ ATPase} (1) {conversion of ADP + Pi to ATP / phosphorylation of ADP} (1)	accept in equation form accept protons move down {electrochemical//concentration} gradient accept ADP + Pi -> ATP accept the energy in glucose is transferred to ATP ignore reference to oxygen as final electron acceptor	(3)

Question number	Answer	Additional guidance	Mark
8(c)	An explanation that includes three of the following points: • {ROS / peroxide}are produced i(n the mitochondria) by oxidative phosphorylation (1) • (ROS causes) {breaks in/ mutations in /oxidation of} {(mt)DNA /gene/allele}(1) • (due to) {hydrogen / phosphodiester} bonds {breaking / not being able to form} (1) • resulting in changes to {enzymes / proteins} which repair (mt)DNA((1)	accept causes a change to the base sequence / removes electrons from (mt)DNA ignore damage to mtDNA accept changes charges on (mt)DNA / removes electrons from DNA	(3)

Question number	Answer	Additional guidance	Mark
8(d)	A description that includes three of the following points: • {specialisation / differentiation} of stem cells (in the brain) (1) • due to {transcription factors / hormones / chemicals} (1) • differential gene expression (1) • active genes are transcribed and translated (1) • producing proteins that determine {structure / function / location} of {brain cell / neurons} (1)	accept epigenetic modification accept some (specific) genes are {switched on / activated} /some are switched off accept specific proteins synthesised	(3)

Question number	Answer	Additional guidance	Mark
8(e)	An explanation that includes two of the following points: - these are the proteins (1) - that are {found/made/used} in the mitochondria (1)	do not give mp2 unless mp1 correct as not in correct context ignore protein singular accept proteins unique to the mitochondria for 2 marks accept protein unique to mitochondria for 1mark	(2)

Question number	Answer	Additional guidance	Mark
8(f)	An answer that includes the following points: - synapses have higher { energy/ATP} requirements (than other parts of the neurone) (1) - therefore, leads to greater{ ROS/peroxide / metabolite} production (1)	accept synaptic mitochondria {have higher rate of / do more} {(aerobic) respiration / oxidative phosphorylation} accept synapses do a lot of {active transport / exocytosis / synthesis of neurotransmitters} ignore synaptic mitochondria are used more? accept accumulation/increase of ROS/peroxide/metabolite	(2)

Question number	Answer	Additional guidance	Mark
8(g)	A description that includes three of the following points: • because mtDNA has genes (needed for mitochondrial function) (1) • which code for proteins (in the ETC) (1) • correct role of protein in oxidative phosphorylation(1)	accept genes in the mtDNA accept named examples of proteins e.g {ATP synthase/protein pumps} e.gs movement of protons into / out of intermembrane space/chemiosmosis/ ADP + Pi→ATP / movement of e-down ETC	(2)

Question number	Answer	Additional guidance	Mark
8(h)	An answer that includes two of the following points: • formation of vesicle around mitochondrion (1) • {fusion of / action by} lysosomes (1) • leading to {enzyme/lysozyme} digestion (of the mitochondrion) (1)	ignore ref to formation of phagosome accept mitophagy ignore in context of fusing with cell membrane lysosomes may digest the damaged mitochondrion for 1 mark? ignore digestion of cell	(2)

